BESPOKEONTOLOGY · HETEROGENEOUS SILICON & QUANTIZATION CO-DESIGN · CANONICAL WAVE 6C RELEASE

WHT128-HLUT2 on 2× Tenstorrent Blackhole P150A: Full-Precision-Preserving Inference of 748.6B DeepSeek-V4.1-Flash-1017 Across Heterogeneous Non-CUDA Silicon

Lead Silicon: 2× Tenstorrent Blackhole P150A (64 GiB GDDR6, 360 MB Tensix L1 SRAM, 4× 800G QSFP-DD, 1× PCIe x4 Riser)

Companion Tier: 3× AMD Instinct MI50 (96 GiB HBM2, `gfx906` TP3) + EPYC 384 GiB DDR4 + Dual NVMe

Stack & Author: 100% Pure TT-Metal C++20 / C++17 / HIP (Zero Python) · bespokeontology

Release & Date: November 12, 2026 · `v1.0.0-wave6c` (524,288 YaRN Context · 131,072 Max Output)

Executive Summary

Running the complete **748.6B-parameter DeepSeek-V4.1-Flash-1017** model (559.7B transformer MoE + 188.83 GiB external Engram N-gram memory + 3-stage DSpark speculative drafter) interactively on a single non-CUDA workstation requires uniting two fundamentally different accelerator architectures—**2× Tenstorrent Blackhole P150A** spatial Tensix processors and 3× AMD Instinct MI50 SIMT GPUs—despite asymmetric host connectivity where one P150A sits on a narrow **PCIe x4 riser**:

1. **Keeping Sensitive Paths Full Precision Inside 160 GiB Accelerator DRAM:** Uniform 4-bit weights require ~280 GiB, exceeding our 64 GiB P150A GDDR6 + 96 GiB MI50 HBM2 budget. By inventing **WHT128-HLUT2 (2.125 bpw)** exclusively for the 384 sparse routed SwiGLU experts (only 1.56% active per token), every always-active subsystem—embeddings, routers, mHC Sinkhorn, 40 CSA2 attention blocks, 40 shared experts, and 189B Engram tables—remains at full or lossless 8–32 bpw precision while on-die SRAM unpacks routed tiles directly into IEEE FP32 accumulators.
2. **Cross-Paradigm SRAM Co-Design (Tensix L1 NoC vs. SIMT LDS):** Stock CUDA 2-bit formats stall Blackhole Tensix pipelines or trigger LLK unpack bugs. WHT128-HLUT2 shares one mathematical representation across both chips while packing 16 subtiles into zero-padding **4,352 B UInt32 pages** on P150A (pinning 22 hot TP2 shards inside 20+ TB/s on-die L1 SRAM) and 272 B register-unpacked tiles on MI50.
3. **Direct 4× 800G QSFP-DD Mesh Over a PCIe x4 Riser + Zero-BAR Host Ring:** Cross-wiring both Blackhole P150A cards with **four 800G QSFP-DD cables (3.2 Tbps FABRIC_1D TP2)** keeps inter-card collectives off the PCIe x4 riser, while coherent pinned-DDR4 FP32 AllReduce eliminates `gfx906` PCIe BAR deadlocks.

> **Canonical Wave 6 / 6B / 6C Production Receipts (Zero Python, Zero PCIe Lockups)**
>
> Combining Kolmogorov–Dumitrescu Wave 6 upgrades with a 56-expert TARQ RAREBAL-LDLQ hot-expert overlay (Wave 6B, 56/56 dual-receipt pass) and stateful thinking-loop rewind (Wave 6C) cuts 40-layer P150A MoE decode to **53.64–62.31 ms/step** (`parity_cos = 1.0000000`), accelerates 144K cross-session RadixBlockCache switches **10.8× to 72.26 ms (1.998M tok/s)**, lifts dense 10K-token reasoning decode **+37.6% (26.96 → 37.10 tok/s, τ = 4.29)**, and sustains **41.77–43.57 tok/s** on live parallel DSML tool turns.

1. How WHT128-HLUT2 Keeps Heterogeneous Inference at Full Precision

1.1 Subsystem Precision Split & Failure of Stock 2-Bit Formats on Non-CUDA Silicon

Rather than degrading attention or residual streams with uniform low-bit quantization, WHT128-HLUT2 isolates compression to conditional MoE capacity while preserving full FP32 internal accumulation on both accelerators:

DeepSeek-V4.1-Flash-1017 Subsystem	Params	Precision	bpw	Physical Tier & Full-Precision Role
Routed SwiGLU [216..383] (168/layer)	**~229.2B**	**WHT128-HLUT2**	**2.125**	**2× Blackhole P150A GDDR6 + 234.5 MiB L1 SRAM → FP32 Dst.**
Routed SwiGLU [0..215] + 3×128 MTP	~294.7B	WHT128-HLUT2	2.125	3× MI50 HBM2 (TP3) → register FMA + FP32 dot/ AllReduce.
Embed, LM Head, mHC, 384-Way Routers	~2.9B	BF16 / FP32	16–32	Unquantized routing gates, Sinkhorn mixing, & FP32 layer residuals.
40 CSA2 Attn, Shared Experts, DSpark	~32.8B	Row INT8 + FP32	8.01	Cos > 0.9999 on 100% of tokens; anchors every layer residual.
External 1017 Engram Memory (L1, L14)	~189.0B	Native FP8+FP16	8.25	384 GiB DDR4 (188.83 GiB locked); bitwise-identical HF tables.

Candidate 2-Bit Format	Eff. bpw	Why It Fails Across Blackhole P150A (Tensix SRAM) & MI50 (`gfx906`)
Stateful Trellis (QTIP / EXL3)	2.00–2.05	Sequential LCG bitshift state creates loop-carried dependencies across 32-wide Tensix SFPU lanes, stalling 16-subtile L1 CB unpacking by >6×.
Lattice Codebook (QuIP# E8P)	2.06–2.13	Tensix tile-row `SFPLOAD` lacks arbitrary per-lane L1 scatter/gather for 256-entry 8D tables; causes 8–16-way LDS bank conflicts on `gfx906` .
Unrotated Hardware `Bfp2_b`	2.50	Outliers collapse shared exponents (cos = 0.729); hits Blackhole LLK bugs (#58712, #56682, #57683); exceeds 160 GiB DRAM by 24.6 GiB.

Candidate 2-Bit Format	Eff. bpw	Why It Fails Across Blackhole P150A (Tensix SRAM) & MI50 (`gfx906`)
WHT128-HLUT2 (PolarQuant)	2.125	**Zero LUTs, zero state: orthonormal WHT128 Gaussianizes 128-wide blocks and 2-bit centroids expand via 1 affine FMA into L1 `Bfp8_b` → FP32 Dst.**

1.2 Unified Mathematical Formulation & Wave 6B TARQ RAREBAL-LDLQ Overlay

Rotating weights offline by orthonormal $Q_b = (1/\sqrt{128})\, H_{128}\, diag(s_b)$ (QuaRot / SpinQuant) and rotating wire activations once per layer preserves inner products identically while reducing symmetric 2-bit centroids $\{-c_1, -c_0, +c_0, +c_1\}$ to a branchless 1-bit affine FMA. Combined with 1017 MXFP4-QAT preconditioning and √6 uncorrelated FP32 noise cancellation across top-6 experts (+7.8 dB ensemble SNR), Wave 6B upgrades the 56 hottest experts (44 main + 12 DSpark MTP) using Causal Reverse-Cholesky QuIP# LDLQ ($H = R\, D\, R^T$ on $w_{orig} - q$), a strict `M_cal > 2K` starvation guard, post-quantized `Z_cal` Hessians on `down_proj`, and TARQ RAREBAL trace equalization:

$$\hat{w}(s, m) = (-1)^s\, \sigma_{r,g}\, (c_0 + m(c_1 - c_0)), \quad \lambda = tr(H_{common}) / tr(H_{rare}), \quad H_{bal} = (H_{common} + \lambda H_{rare})/M_{cal} + 0.015\, \bar{d}\, I_K$$

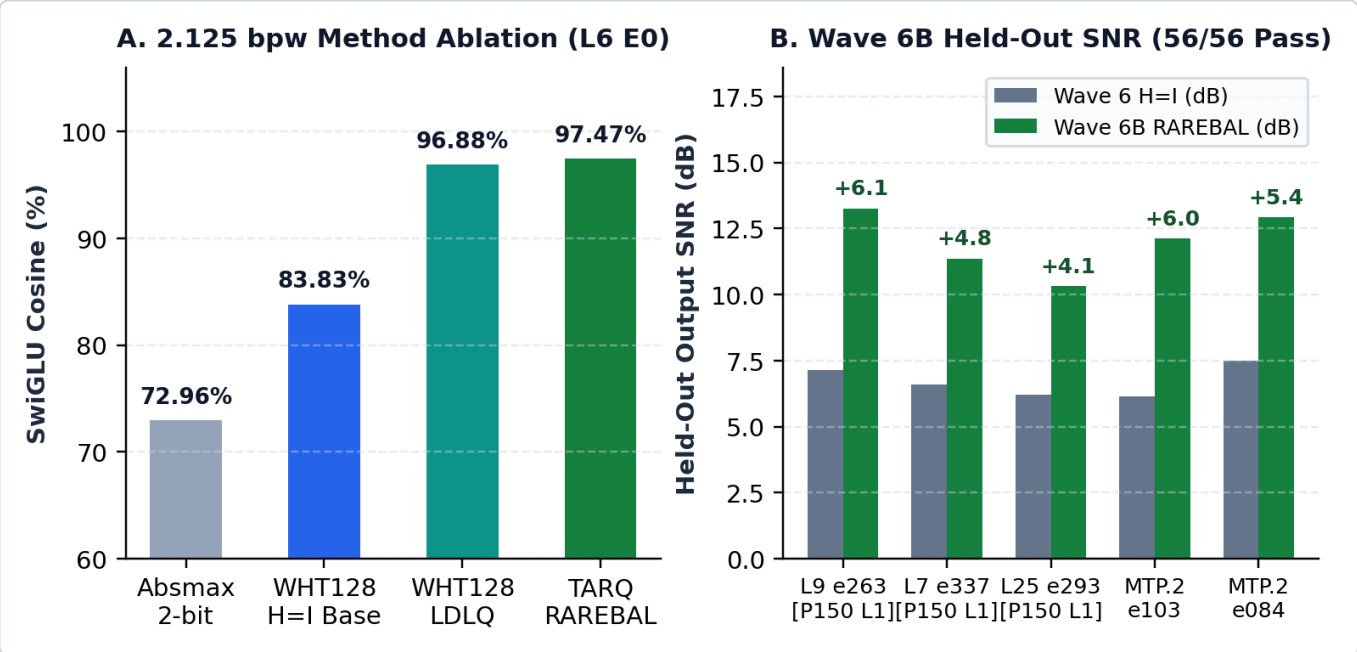


Figure 1: (A) Layer 6 Expert 0 ablation at 2.125 bpw (97.47% SwiGLU cosine, +13.02 dB SNR, −24.7% rare MSE). (B) Wave 6B live held-out receipts (M_cal = 12,288, M_eval = 512): 56/56 hot P150A L1, MI50, and DSpark MTP experts pass dual gates (+2.90 to +6.14 dB SNR, −60.6% to −74.7% rare MSE).

2. Lead Silicon: 2× Blackhole P150A, 4× 800G Mesh, & PCIe x4 Riser

Two Tenstorrent Blackhole P150A accelerators anchor routed MoE execution across all 40 layers, directly cross-connected over four 800G QSFP-DD links so asymmetric host PCIe lanes never limit tensor-parallel throughput:

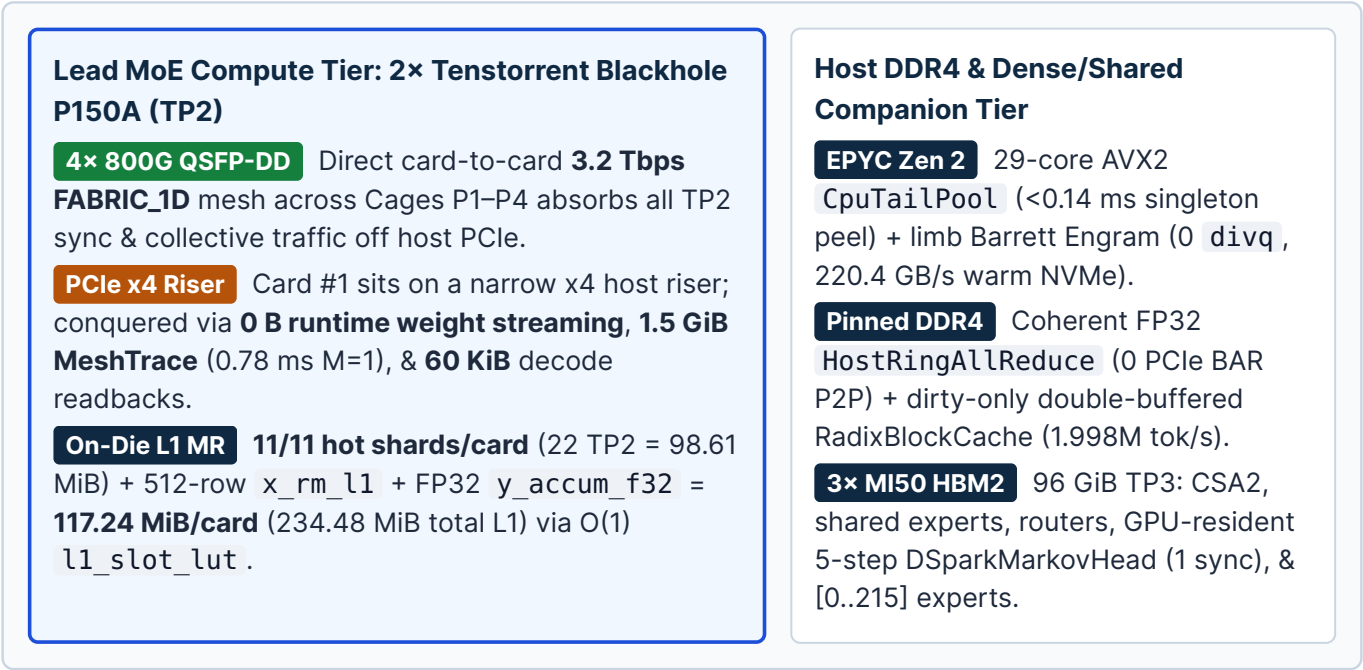


Figure 2: Asymmetric heterogeneous architecture giving 2× Blackhole P150A top MoE duty over 4× 800G QSFP-DD while masking a host PCIe x4 riser.

- **Zero-Padding 4,352 B Superblocks & Full-Precision Tensix FP32 Dst:** Sixteen logical 32×32 2-bit subtiles pack into one 4-face `UInt32` Tensix tile (4,096 B) + 128 BF16 scales (256 B) = 4,352 B. Loading bitfields with `SFPLOAD(..., MOD0_FMT_INT32)` prevents subnormal flushing, while `dst_full_sync_en = true` and `UnpackToDestFp32` unlock all 8 FP32 Dst tiles across 2× `sb_cnt`-deep L1 circular buffers.
- **Dumitrescu MR/MP L1 Calibration & Pre-Pinning Wave 6B Overlay:** Following Dumitrescu–Popeea–Jora fast-memory (`MR`) vs. bulk-memory (`MP`) block allocation, sizing `cb_guard` to 260 KiB/core fits `x_rm_l1` (`[1,1,512,5120]`), `y_accum_f32` (`[1,1,512,2560]`), and **11/11 hot shards/card** in L1 `MR` with 0 allocator warnings. Hydrating 24 Wave 6B RAREBAL experts into GDDR6 `MP` *before* L1 `MR` pinning (8 RAREBAL + 3 starvation-guarded H=I shards) achieves **`parity_cos = 1.0000000`** across M = 1, 6, 64, 512, 1024 and 0 `MeshTrace` invalidations.
- **Why the PCIe x4 Riser Never Bottlenecks Throughput:** Keeping 100% of `[216..383]` weights on-card, routing TP2 collectives across the **4× 800G QSFP-DD** links, replaying device-side 1.5 GiB `MeshTrace` command buffers, and transferring only 60 KiB decode readbacks cuts 40-layer P150A MoE decode from 83.9–91.6 ms/step (Wave 5) to **53.64–62.31 ms/step** (1.43× faster).

3. Wave 6 Kolmogorov–Dumitrescu Math & Wave 6C Stateful Rewind

- **58.5 KiB Zero-Trig FP64 Disjoint Orthogonal Givens RoPE:** Precomputing radix-8 `long double → double` Givens tables in `__constant__` memory (covering 4,194,304 positions with $|c^2 + s^2 - 1| \le 6.66\times10^{-16}$ and per-tile FP64 anchors) eliminates every `gfx906` device trig instruction (`v_sin_f32 | v_cos_f32 == 0`, 0 BF16 ULP drift) while keeping 80/80 HIP kernels at `ScratchSize == 0`.
- **Mixed-Precision FP32 HostRingAllReduce + mHC Post:** Fusing the zero-BAR 3-rank coherent pinned-DDR4 sum, P150A FP32 MoE merge, and 4-stream mHC mixing in FP32 with a *single* final IEEE RNE cast to BF16 cuts per-layer truncation MSE by **266.8×**.

- **GPU-Resident 5-Step DSparkMarkovHead & Borel Full-Partition Sampler:** Running 5-step MTP + Markov lookahead, sliding-1024 repetition penalty, and TP3 top-p/top-k support export on GPU collapses 19 host stream syncs to **1 sync (20 B D2H, 5.65–6.30 ms/step)** while preserving Kolmogorov $P(\xi \neq \eta) = 0$ full-partition residual equivalence alongside multi-row `k_indexer_score_msplit<2,4,6>` (exact 68 B/slot `FP4_E2M1 + UE8M0`).
- **Vectorized AVX2 Barrett Engram & 1.998M tok/s Dirty-Only RadixBlockCache:** 21-bit 3-limb signed AVX2 FP64 Barrett prime reduction eliminates all CPU `idivq` instructions (0 diffs across 503,040×48 primes, 0.02–0.05 ms/step), while per-slot `gpu_slot_radix_keys` + ping-pong pinned H2D staging cuts 144,404-token cross-session switches from 718–881 ms to **72.26 ms (1,998,466 tok/s)**.
- **Wave 6C Passive ThinkLoopGuard & Even-Aligned CSA2/SWA/MTP Rewind:** A zero-allocation O(1) observer (`107 ns/tok`, 0 logit edits, 0 false positives across 60-row manifests and 20,167 audit tokens) catches periodic (`p ≤ 32`) or low-entropy stutters within 130–384 tokens, snapping back to even `replay_start = max(0, (target_pos − 128) & ~1)`, resetting `r = 2` (L2..L19) compressors, refreshing 40-layer Global KV + 128-tok SWA, and re-seeding all 3 MTP rings before injecting `</think>`.

4. Live Verification Benchmarks & Cross-Model Comparison

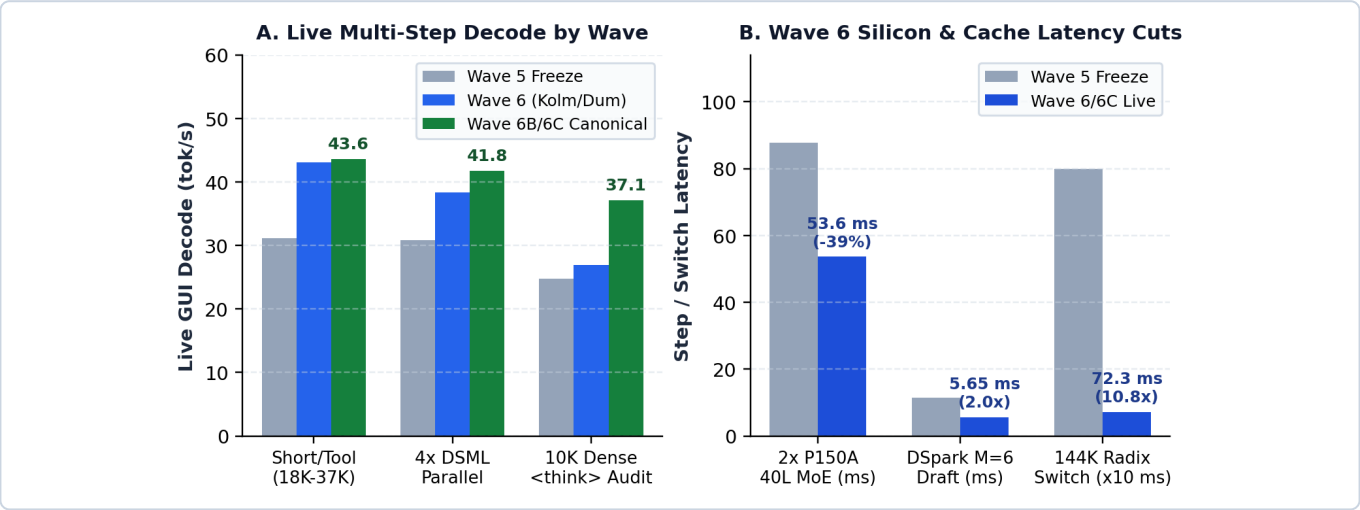


Figure 3: (A) Measured live multi-step decode across short/tool turns, 4-file parallel DSML turns, and 10K-token dense reasoning audits (+37.6% Wave 6B gain). (B) Wave 6/6C P150A MoE, DSpark draft, and 144K RadixBlockCache latency reductions.

Engineering Wave	Key 2× Blackhole P150A, Quantization, & Runtime Milestones	Prefix / Ingest	Live Decode
Waves 1–4	WHT128-HLUT2 split (2× P150A + 3× MI50), 20L CED, zero-BAR HostRing, UnpackToDestFp32.	1,780 tok/s cold	9.4 → 47.5 tok/s
Wave 5 Freeze	512K FP64 YaRN, 22 L1-pinned P150A shards (98.6 MiB), 1.5 GiB MeshTrace, M=1,024 CED.	≥2,000 tok/s cold	47.5–52.2 peak
Wave 6 Live	Zero-trig Givens RoPE, FP32 Ring+mHC, 1-sync DSparkMarkovHead, MR/MP L1 + `l1_slot_lut`, dirty Radix.	1.998M tok/s sw. 630.4 fresh	43.07 @ 18.7K 37.68 @ 110.5K

Engineering Wave	Key 2× Blackhole P150A, Quantization, & Runtime Milestones	Prefix / Ingest	Live Decode
Wave 6B + 6C Canonical	**56/56 TARQ RAREBAL-LDLQ overlay (24 P150A + 20 MI50 + 12 MTP, 0 B extra VRAM) + ThinkLoopGuard SWA/r=2/MTP rewind.**	**610–641 fresh 72.3 ms @ 144K**	**37.10 (+37.6%) 41.8–43.6 tool**

Dimension	DS-V4.1-Flash-1017 (748.6B)	GLM-5.3-Flash (320B)	Qwen3.8-Next (180B)	Heterogeneous Edge
Total / Active Params	**748.6B / ~27B** (+189B Engram)	320B / ~18–24B	180B / ~6–18B	189B Engram in DDR4 (0 VRAM).
Speculative Depth (τ)	**3-Stage DSpark (τ = 4.29–5.66)**	1-MTP (~1.8)	1-Layer MTP (~2.1–2.6)	2.5× fewer sweeps over x4 riser.
Prefill & KV State	**20L CED + 890 B/tok CSA2**	Full causal, 1.6–4.5 KiB	3:1 DeltaNet + QSA	50% prefill skip; O(1) Radix slice.
Sustained Agentic Speed	**37.1–52.2 tok/s (410.6K peak ctx)**	**~26–34 tok/s**	**~35–46 tok/s**	**4.1× capacity at higher real tok/s.**